

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

$$\sin 2A = 2 \sin A \cos A \quad \text{or} \quad \frac{2 \tan A}{1 + \tan^2 A}$$

$$\cos 2A = \cos^2 A - \sin^2 A \quad \text{or} \quad \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$2 \cos^2 A - 1$$

$$1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

$$2 \sin A \cos B = \sin(A+B) + \sin(A-B)$$

$$2 \cos A \sin B = \sin(A+B) - \sin(A-B)$$

$$2 \cos A \cos B = \cos(A+B) + \cos(A-B)$$

$$2 \sin A \sin B = \cos(A-B) - \cos(A+B)$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\cos^2 A = \frac{1 + \cos 2A}{2}$$

$$\sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$$

$$\sin C - \sin D = 2 \cos \left(\frac{C+D}{2} \right) \sin \left(\frac{C-D}{2} \right)$$

$$\cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$$

$$\cos C - \cos D = 2 \sin \left(\frac{C+D}{2} \right) \sin \left(\frac{D-C}{2} \right)$$

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Delta = \frac{1}{2} ac \sin B$$

$$a = b \cos C + c \cos B$$

$$\sin \theta = \sin \alpha \quad \theta = n\pi + (-1)^n \alpha$$

$$\cos \theta = \cos \alpha \quad \theta = 2n\pi \pm \alpha$$

$$\tan \theta = \tan \alpha \quad \theta = n\pi + \alpha$$

$$\cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}$$

$$\sin \frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{ac}}$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$